DEVELOPMENT OF A SAFETY CHECKLIST FOR CHEST DRAIN INSERTION

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Introduction:

- The 2008 UK National Patient Safety Agency (NPSA) issued a report that 12 deaths and 15 cases of serious harm had between reported between 2005 to 2008, relating directly to chest drain insertion and a total of 2,152 events harm relating to chest drains[1].
- In 2014 the Royal College of Emergency Medicine issued a safety alert following the death of a patient due to chest drain insertion into the incorrect side. Contributing factors included failure to examine the patient, an incorrectly labelled chest x-ray and failure to review the chest X-ray[2].
- Contributing factors to Serious Harm Events and Never Events relating to chest drain insertion include: site of insertion (both rib level and AP positioning), inadequate imaging, excessive insertion of dilator as well as inadequate supervision. Other factors include: anatomical anomalies, patient clinical condition and low level of experience of the doctor performing the chest drain[1,3].
- The use of the WHO surgical safety checklist showed a reduction in post-operative error rates by 36% and thus enhanced safety[4].
- Dartford and Gravesham NHS trust introduced a chest drain insertion checklist in 2011 and reported a reduction in adverse incidents since implementation as well as improved confidence in the procedure amongst trainee staff[5].

Aim: To design a chest drain insertion checklist based on evidence and experience, thereby reducing the non-technical burden and increasing successful insertion rates.

Methods:

- Following a systematic review of current literature, an evidence based checklist was developed (Figure 1).
- Via mixed methodology, including a Delphi process, this was reviewed and commented on by 32 EM consultants in the West Midlands region of England (Figure 2).
- The design and content of the primary checklist was evaluated. We also sought the views of the ED consultants as to where the common errors in chest drain insertion occur.
- The resultant checklist was discussed to saturation within focus groups including middle-grade doctors.
- A final analysis of the data collected was used to update the checklist (Figure 3).

Results:

- Our work indicated that most errors were felt to be related to site of insertion and experience/competence of the doctor.
- Through a re-iterative process we produced a final checklist design that was clinically relevant, user friendly and acceptable to doctors.
- Common errors, as reported by the consultants, were compared with that of the current literature-reported errors and concordance was found.
- Errors included: lack of experience, poor pain relief, incorrect drain insertion (too far or not enough) and insertion outside of the 'Triangle of Safety'.

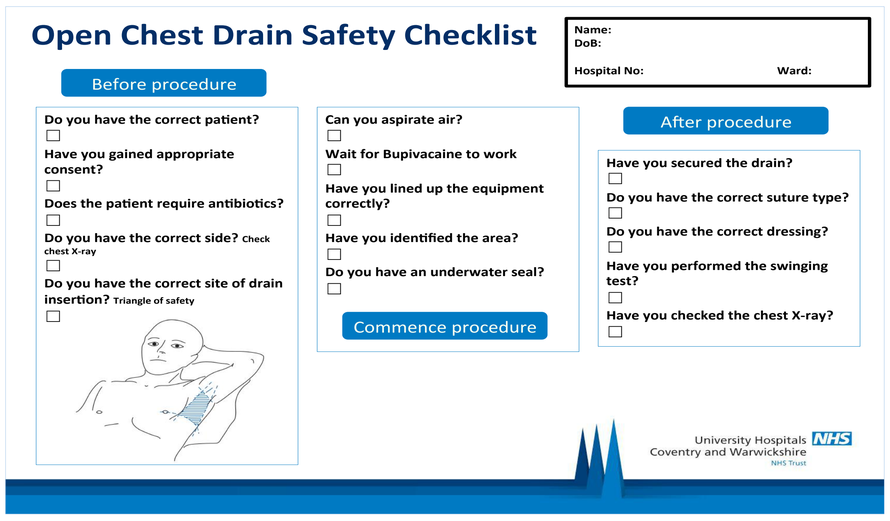
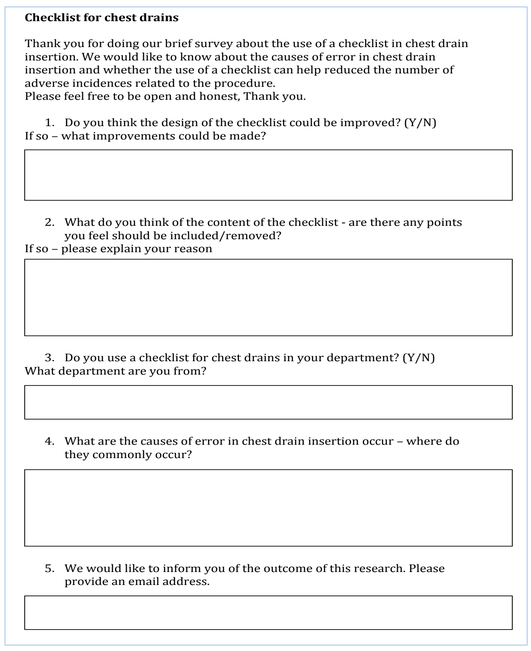


Figure 1: Primary Checklist Design



Checklist for chest drains

Thank you for doing our brief survey about the use of a checklist in chest drain insertion. We would like to know about the causes of error in chest drain insertion and whether the use of a checklist can help reduced the number of adverse incidences related to the procedure.
Please feel free to be open and honest, Thank you.

1. Do you think the design of the checklist could be improved? (Y/N)
If so – what improvements could be made?

2. What do you think of the content of the checklist - are there any points you feel should be included/removed?
If so – please explain your reason

3. Do you use a checklist for chest drains in your department? (Y/N)
What department are you from?

4. What are the causes of error in chest drain insertion occur – where do they commonly occur?

5. We would like to inform you of the outcome of this research. Please provide an email address.

Figure 2: Questionnaire to evaluate checklist

Discussion:

- Checklists have been shown to reduce avoidable complications, as evidenced by the success of the WHO surgical checklist.
- Our study results are in line with the NPSA report, but we also identify stages in the procedure where further checks would be of benefit[1,3].
- 40% of Consultants believed that the insertion of a chest drain into the 'wrong site' was a common error.
- We designed a checklist to help reduce errors associated with the insertion of chest drains that is compatible with both types of chest drain techniques.
- We identified non-technical skills as the area most associated with error and designed the checklist to address this. The theory is that reduced cognitive burden relating to non-technical tasks allows more focus on the technical parts of the procedure.
- We enhanced support at these stages, where error rates are at their greatest, by including other team members via a call and response system.
- Other perceived advantages include: improved communication between staff, enhanced safety, comfort to patients and reduced complications post-procedure.
- Our checklist serves to prompt clinician behaviour, providing an additional level of safety at key points.

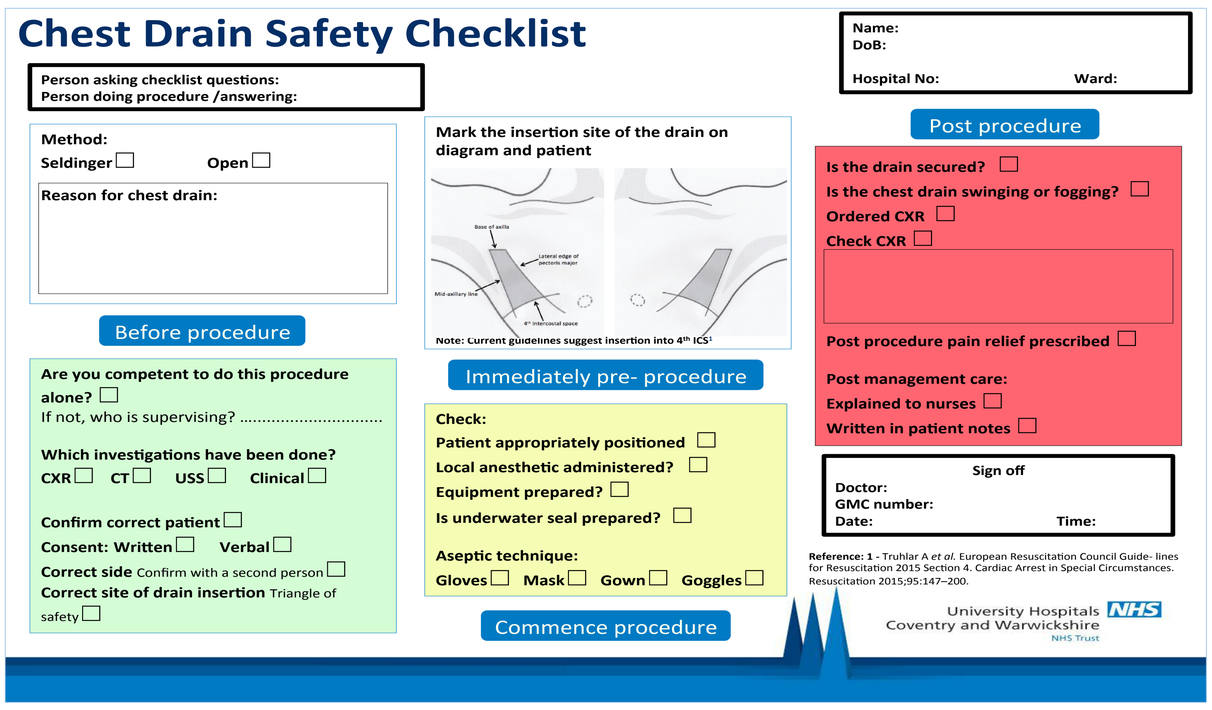


Figure 3: Final Checklist Design

Conclusion:

- A systematic search of current literature highlighted the avoidable harms associated with chest drain insertion.
- This evidence based approach in combination with the experience of 32 ED consultants led to the development of a checklist designed to address the non-technical skill aspects of chest drain insertion.
- The common errors associated with chest drain insertion were also identified, and the check list was designed with a view to help reduce recurrence of these errors.
- The checklist will potentially reduce some of the cognitive burden associated with performing this practical procedure, reducing the overall error/complication rate.
- Further investigations are required to test the effectiveness of the checklist and whether any improvements can be made.

Future Work:

- Implementation of the safety checklist within the clinical environment to determine whether or not the use of such a checklist could help reduce the number of errors in clinical practice.
- Determine the effectiveness of the variation in chest drain insertion and removal, in particular methods of securing the chest drain.

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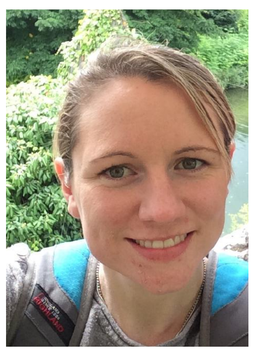

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